

Mario Ayala

Mario Antonio Ayala Valenzuela

Postdoctoral researcher, Chair for Analysis and Modelling

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Research interests

Probability theory, and more precisely scaling limits of interacting particle systems: hydrodynamic limits, fluctuation fields and their SPDE limits, duality and orthogonal duality polynomials, Dirichlet forms and Mosco convergence, condensation. Effective dynamics and coarse-graining of stochastic systems, including the interpretation of multiplicative noise. Stochastic population models.

Positions

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| since 08/2022 | Postdoctoral researcher, Chair for Analysis and Modelling, Technische Universität München, Germany. Host: Johannes Zimmer. |
| 05/2021–07/2022 | Postdoctoral researcher, BioSP, INRAE, Avignon, France. With Jérôme Coville and Raphaël Forien. |

Education

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| 2021 | PhD in Applied Mathematics, Delft University of Technology, the Netherlands. Thesis: <i>Fluctuations for interacting particle systems with duality</i> , defended on 18 February 2021. Promotor Frank Redig, copromotor Gioia Carinci. |
| 2016 | MSc in Applied Mathematics, specialization in Financial Engineering, Delft University of Technology. Thesis: <i>Hydrodynamic limit for the symmetric inclusion process</i> . |
| 2011 | BSc in Mathematics, Facultad de Ciencias, Universidad de Colima, Mexico. Thesis: <i>Estimation of the basic reproductive number for HLB in the state of Colima, Mexico</i> . |

Distinctions and scholarships

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| 01/2022–12/2028 | National Researcher, Level I, Mexican National System of Researchers (SNII). |
| PhD | CONACYT scholarship for PhD studies at Delft University of Technology. |
| MSc | CONACYT scholarship for MSc studies at Delft University of Technology. |
| 2003 | Bronze medal, 17th Mexican Mathematical Olympiad, Guanajuato. |
| 2003 | First place, national Mathematical Skills Contest, ITESO, Jalisco. |

Publications

Published.

- [1] M. Ayala and J. Zimmer. Hydrodynamic limits and non-equilibrium fluctuations for the symmetric inclusion process with long jumps. *Annales Henri Poincaré* **27**(7), 2685–2721 (2026).
- [2] M. Ayala, J. Coville and R. Forien. A measure-valued stochastic model for vector-borne viruses. *ALEA Latin American Journal of Probability and Mathematical Statistics* **21**, 1145–1193 (2024).
- [3] M. Ayala, G. Carinci and F. Redig. Condensation of SIP particles and sticky Brownian motion. *Journal of Statistical Physics* **183**(3), art. 40 (2021).

- [4] M. Ayala, G. Carinci and F. Redig. Higher order fluctuation fields and orthogonal duality polynomials. *Electronic Journal of Probability* **26**, 1–35 (2021).
- [5] M. Ayala, G. Carinci and F. Redig. Quantitative Boltzmann-Gibbs principles via orthogonal polynomial duality. *Journal of Statistical Physics* **171**(6), 980–999 (2018).

Submitted.

- [6] M. Ayala and J. Zimmer. Computing extinction barriers in a quorum-sensing reaction network. [arXiv:2609.39946](#) (2026).
- [7] M. Ayala and B. Vallejo Jiménez. Consumption-investment under alternative stochastic-integral conventions. [arXiv:2602.08527](#) (2026).
- [8] M. Ayala and D. R. M. Renger. Hydrodynamic limits of collisions and fluxes in the exclusion process. [arXiv:2512.20412](#) (2025).
- [9] M. Ayala, N. Dirr, G. A. Pavliotis and J. Zimmer. Reversibility, covariance and coarse-graining for Langevin dynamics: on the choice of multiplicative noise. [arXiv:2511.03347](#) (2025).
- [10] M. Ayala. Mosco convergence of independent particles and applications to particle systems with self-duality. [arXiv:2406.19088](#) (2024).
- [11] M. Ayala, J. Coville and S. Soubeyrand. Group dispersal modelling revisited. [arXiv:2405.08384](#) (2024).

In preparation.

- [12] M. Ayala, N. Dirr and J. Zimmer. Atomic limits of empirical measures of interacting particle systems with self-duality.
- [13] M. Ayala and F. Redig. Duality for SPDEs.
- [14] M. Ayala and J. Zimmer. Cooperation through a shared environment: hydrodynamic limits and large deviations for cheater invasion.
- [15] M. Ayala and J. Zimmer. Derivation of the symmetric exclusion process from a monitored Bose-Hubbard model.
- [16] M. Ayala and D. R. M. Renger. Cumulants, SEP and criticality.

Theses and technical reports.

- M. Ayala. Fluctuations for interacting particle systems with duality. PhD thesis, Delft University of Technology (2021).
- M. Ayala. Hydrodynamic limit for the symmetric inclusion process. MSc thesis, Delft University of Technology (2016).
- Evaluation of *Aedes aegypti* control campaigns. Technical report for the Ministry of Health of the State of Colima, Mexico (2009).
- M. Ayala, C. L. Pawling, A. N. Smith, L. Gao, D. Hiebeler and B. R. Morin. An epidemiological approach to the dynamics of chytridiomycosis on a harlequin frog population. Technical report, Mathematical and Theoretical Biology Institute, Arizona State University (2006).

Talks

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| 07/2024 | <i>Scaling limits for the inclusion process with long jumps.</i> University of Modena and Reggio Emilia, Italy. |
| 04/2023 | <i>Condensation of SIP particles and sticky Brownian motion.</i> Workshop “In search of model structures for non-equilibrium systems”, Münster, Germany. |

03/2023	<i>Fluctuation fields and orthogonal self-dualities</i> . Seminário de Probabilidade e Mecânica Estatística, online.
11/2021	<i>A spatial stochastic model for a vector-borne virus population: some scaling limits</i> . Séminaire Mathématiques, Évolution, Biologie, Aix-Marseille Université, France.
08/2021	<i>A spatial stochastic model for a vector-borne virus population</i> . YEP XVII: Interacting particle systems, Eurandom, Eindhoven, the Netherlands.
07/2021	<i>Higher order fluctuation fields and orthogonal duality polynomials</i> . Encontro Nacional da SPM 2021, session on stochastic duality for Markov processes, online.
02/2020	<i>Higher order fluctuation fields and orthogonal duality polynomials</i> . Third Haifa Probability School, Workshop on Random Geometry and Stochastic Analysis, Technion, Haifa, Israel.
05/2019	<i>Condensation of SIP particles and sticky Brownian motion</i> . 27th Meeting of PhD students in Stochastics in the Netherlands.
05/2019	<i>Condensation of SIP particles and sticky Brownian motion</i> . NDNS+ PhD Days 2019, the Netherlands.
02/2018	<i>Sistemas de partículas que interactúan y aplicaciones a modelos de distribución de la riqueza</i> . Internal seminar, Facultad de Economía, Universidad de Colima, Mexico.
08/2006	<i>An epidemiological approach to the dynamics of chytridiomycosis in a harlequin frog population</i> . SIAM Conference on the Life Sciences, Raleigh, USA.
<i>Posters.</i>	
07/2019	<i>Condensation of SIP particles and sticky Brownian motion</i> . LMS Research School “Random structures: from the discrete to the continuous”, University of Bath, United Kingdom.
10/2006	<i>An epidemiological approach to the dynamics of chytridiomycosis in a harlequin frog population</i> . SACNAS National Conference, Tampa, USA.

Schools and conferences attended

02/2020	Third Haifa Probability School, Workshop on Random Geometry and Stochastic Analysis, Technion, Haifa, Israel.
07/2019	LMS Research School “Random structures: from the discrete to the continuous”, University of Bath, United Kingdom.
07/2018	CIMPA School “Geometry and scaling of random structures”, Universidad de Buenos Aires, Argentina.
04/2017	Pre-school “Stochastic dynamics out of equilibrium”, CIRM, Marseille, France.
09/2016	VIII Escuela de Probabilidad y Procesos Estocásticos, CIMAT, Guanajuato, Mexico.
09/2015	7th General AMaMeF and Swissquote Conference, Lausanne, Switzerland.
summer 2006	Mathematical and Theoretical Biology Institute (MTBI), Arizona State University, USA.

Teaching

WS 2024/25	Exercise coordinator, <i>Functional Analysis</i> , Technische Universität München. Coordination of the tutors, design of problem sheets and exams; more than 240 students.
SS 2023, SS 2024	Exercise coordinator, <i>Calculus III: Integral and Differential Calculus</i> , Technische Universität München. Coordination of the tutors, design of problem sheets and exams; more than 580 students.
WS 2022/23, WS 2023/24	Tutor, <i>Functional Analysis</i> , Technische Universität München.
2019–2020	Teaching assistant, Delft University of Technology: <i>Stochastic Differential Equations</i> (spring 2020), <i>Linear Algebra</i> (winter 2019/20), <i>Analyse en differentiaalvergelijkingen</i> (autumn 2019).
08–12/2018	Lecturer, <i>Numerical Methods</i> (in Spanish), Universidad de Colima, Mexico.
since 2005	Trainer for the Mexican Mathematical Olympiad in Colima, Mexico, on and off: regularly in 2005–2010, and online since 2021.

Service

Referee	<i>The Annals of Probability</i> , <i>The Annals of Applied Probability</i> , <i>Electronic Journal of Probability</i> , <i>Probability and Mathematical Physics</i> , <i>Journal of Statistical Physics</i> , <i>Proceedings of the Royal Society A</i> , <i>Brazilian Journal of Probability and Statistics</i> , <i>Markov Processes and Related Fields</i> .
11/2022	Mentor of the TUM team “Quod erat demonstranTUM” at the International Student Team Competition in Mathematics (ISTCiM 2022), University of Silesia, Katowice, Poland: bronze medal and third prize.

Other professional experience

07–08/2017	Consultant, Dubai Community Development Authority and Dubai Airports Authority, United Arab Emirates.
03–07/2016	Intern, Dutch State Treasury Agency, Ministry of Finance, the Netherlands. Analysis of Dutch inflation-linked bonds.
2007–2014	Administrative support and operations, Ministry of Health of the State of Colima, Mexico.

Languages and computing

Languages	Spanish (native), English (fluent), German (A2).
Computing	Python, R, MATLAB, $\text{\LaTeX}$.